

Chapter 4:

5. a. $8 \% 5 = 3$

b. $-8 \% 5 = -3, 2$

c. $8 \% -5 = 3, -2$

d. $-8 \% -5 = -3, 2$

7. The algorithm for computing the UPC check digit does not work when using $10 - (\text{total} \% 10)$ instead of using $9 - ((\text{total} - 1) \% 10)$ because when any number that is a multiple of 10 is inputted, you will get 0 from the first expression and 1 from the second expression.

8. The program would still work if the expression $9 - ((\text{total} - 1) \% 10)$ were replaced by $(10 - (\text{total} \% 10) \% 10)$

11. a. 0 2

b. 4 11 6

c. 0 8 7

d. 3 4 5 4

12. a. 6 16

b. 6 -7

c. 6 23

d. 6 15

Chapter 5:

1. a. 1

b. 1

c. 1

d. 0

4. `i = i == j ? 0 : (n > j ? 1 : -1);`

5. The following if statement “if (n >= 1 <= 10) is legal, but whenever a number is inputted including 0 the statement “n is between 1 and 10” is printed to the terminal.

11. file “areaCode.c” included in zip

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    int num;
```

```
    printf("Enter a area code: ");
```

```
    scanf("%d", &num);
```

```
    switch(num)
```

```
    {
```

```
        case 229: printf("Albany"); break;
```

```
        case 404: printf("Atlanta"); break;
```

```
        case 470: printf("Atlanta"); break;
```

```
        case 478: printf("Macon"); break;
```

```
        case 678: printf("Atlanta"); break;
```

```
        case 706: printf("Columbus"); break;
```

```
        case 762: printf("Columbs"); break;
```

```

        case 770: printf("Atlanta"); break;

        case 912: printf("Savannah"); break;

        default: printf("Area code not recognized"); break;
    }

}

```

Chapter 5, Programming Project 6: file “UPC.c” included in zip

```

#include <stdio.h>

int main()
{
    int d, lastD, i1, i2, i3, i4, i5, j1, j2, j3, j4, j5, first_sum, second_sum, total, final;

    printf("Enter the first (single) digit: ");
    scanf("%1d", &d);
    printf("Enter the last (single) digit: ");
    scanf("%1d", &lastD);
    printf("Enter first group of five digits: ");
    scanf("%1d%1d%1d%1d%1d", &i1, &i2, &i3, &i4, &i5);
    printf("Enter second group of five digits: ");
    scanf("%1d%1d%1d%1d%1d", &j1, &j2, &j3, &j4, &j5);

    first_sum = d + i2 + i4 + j1 + j3 + j5;
    second_sum = i1 + i3 + i5 + j2 + j4;

```

```
total = (3 * first_sum) + second_sum;
```

```
printf("Check digit: ");
```

```
final = (10 - (total % 10)) % 10;
```

```
final = lastD == final;
```

```
switch (final)
```

```
{
```

```
case 1:
```

```
    printf("VALID\n");
```

```
    break;
```

```
default:
```

```
    printf("NOT VALID\n");
```

```
    break;
```

```
}
```

```
return 0;
```

```
}
```